Sales Insights Dashboard

– An Interactive Business Intelligence Report Using Power BI

The Sales Insights Dashboard is an interactive Power BI solution designed to evaluate sales performance, revenue trends, profitability, and customer segmentation.

It converts raw transactional data into meaningful insights through KPIs, visualizations, filters, and detailed analytical views. This enables decision-makers to monitor business performance effectively and uncover growth opportunities.

Objectives

The key objectives of this project include:

- Analyzing overall sales and profit performance
- Tracking monthly revenue, profit, and order trends
- Comparing sales contributions across product categories
- Understanding customer segment distribution
- Evaluating the impact of discounts on profitability
- Identifying top-performing and underperforming products
- Enabling interactive filtering for data-driven decision-making

Tools & Technologies Used

- **Power BI Desktop** – Dashboard development and visualization
- **DAX (Data Analysis Expressions)** – Measures and calculations
- **Microsoft Excel** – Data source
- **Bookmarks & Buttons** – Navigation across report pages
- **Conditional Formatting** – Visual KPI indicators

Dataset Description

The dataset consists of transactional sales records with the following fields:

- Order ID
- Order Date
- Customer Name
- Product Name
- Category
- Sub-Category

- Region
- Segment
- Sales
- Profit
- Quantity
- Discount

Data Preparation Steps

The dataset was preprocessed using the following steps:

- Imported the Excel dataset into Power BI
- Handled missing or null values
- Converted the Order Date field into a proper date format
- Created a Month-Year field for trend analysis
- Established data relationships where necessary
- Developed calculated measures using DAX

DAX Measures Used

1. Basic KPIs

- Total Revenue
- Total Profit
- Total Orders

2. Time Intelligence

- Last Month Revenue
- Last Month Profit
- Last Month Orders

3. Growth Indicators

- Revenue Change (%)
- Profit Change (%)
- Orders Change (%)

4. Additional Measures

- Profit Margin
- Average Discount

Dashboard Structure

The dashboard is divided into **three interactive pages**, each serving a distinct analytical purpose.

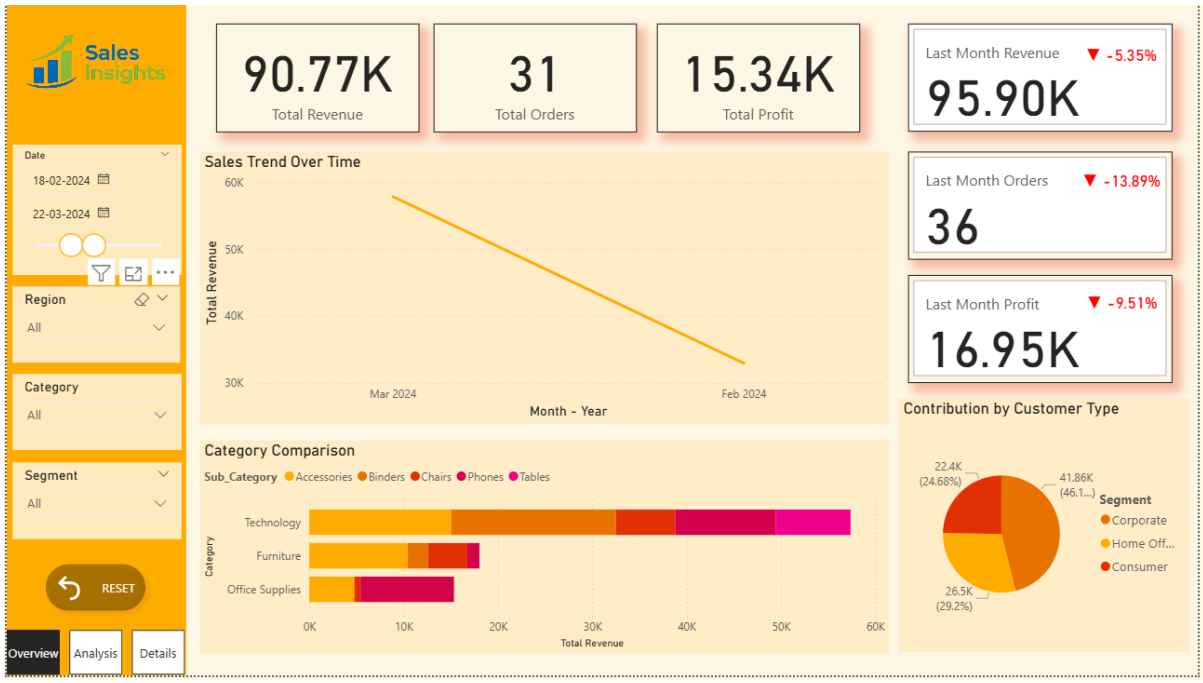
Page 1: Overview

This page presents a high-level snapshot of business performance.

Visuals Included:

- KPI Cards: Revenue, Orders, Profit
- Last Month Comparison Cards
- Sales Trend Over Time (Line Chart)
- Category-wise Comparison (Stacked Bar Chart)
- Customer Segment Contribution (Pie Chart)

Purpose:
To provide a quick and clear understanding of overall business performance.



Page 2: Analysis

This page offers deeper analytical insights into performance drivers.

Visuals Included:

- Profit Contribution by Category (Waterfall Chart)
- Revenue by Region (Map Visualization)
- Profit vs. Discount Analysis (Scatter Plot)
- Sales Conversion Funnel

Purpose:
To identify the factors influencing business performance trends.

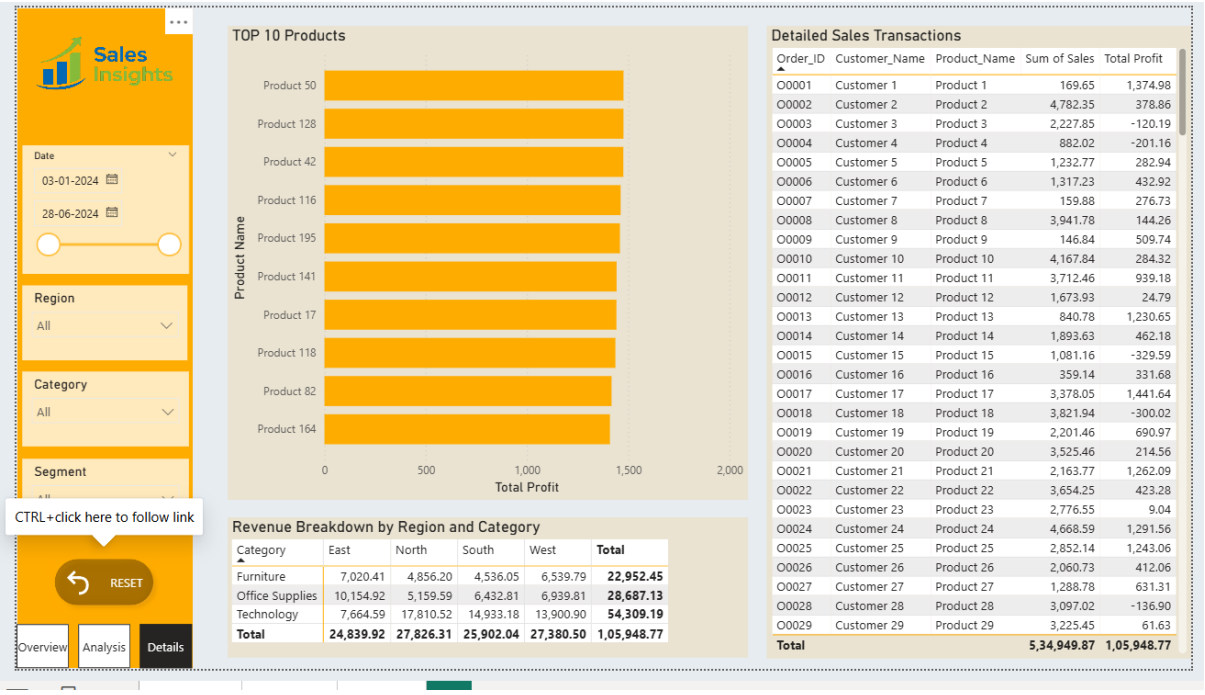
Page 3: Details

This page focuses on granular, transaction-level insights.

Visuals Included:

- Top 10 Products by Profit
- Detailed Sales Transactions (Table)
- Revenue Breakdown by Region and Category (Matrix)

Purpose:
To enable in-depth exploration of detailed business data.



Interactive Features

The dashboard includes several interactive elements to enhance usability:

- Date slicer
- Region slicer
- Category slicer
- Segment slicer
- Reset button
- Multi-page navigation buttons
- Dynamic KPI indicators (▲ ▼ with color coding for performance trends)

Key Business Insights

- The Technology category generates the highest revenue
- Discounts can significantly reduce profit margins in some cases
- Corporate customers contribute a substantial share of revenue
- A small number of products account for a large portion of profits
- Some transactions result in losses despite generating sales
- Monthly revenue trends show noticeable fluctuations

11. Conclusion

The Sales Insights Dashboard effectively transforms raw sales data into actionable business intelligence. It empowers users to track KPIs, analyze trends, assess profitability, and make informed strategic decisions.

This project highlights strong capabilities in:

- Data visualization
- Dashboard design
- Business analysis
- DAX calculations
- Interactive reporting

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12. Future Enhancements

- Incorporate sales forecasting models
- Enable real-time data integration
- Implement customer churn prediction
- Add AI-driven recommendations
- Automate report distribution via email

Developed By

Aarsi Singh
Roll no:2328059